When Rough Seas Become Deadlier

Migrant mortality and the 2017 Italy–Libya Memorandum on the Central Mediterranean Route

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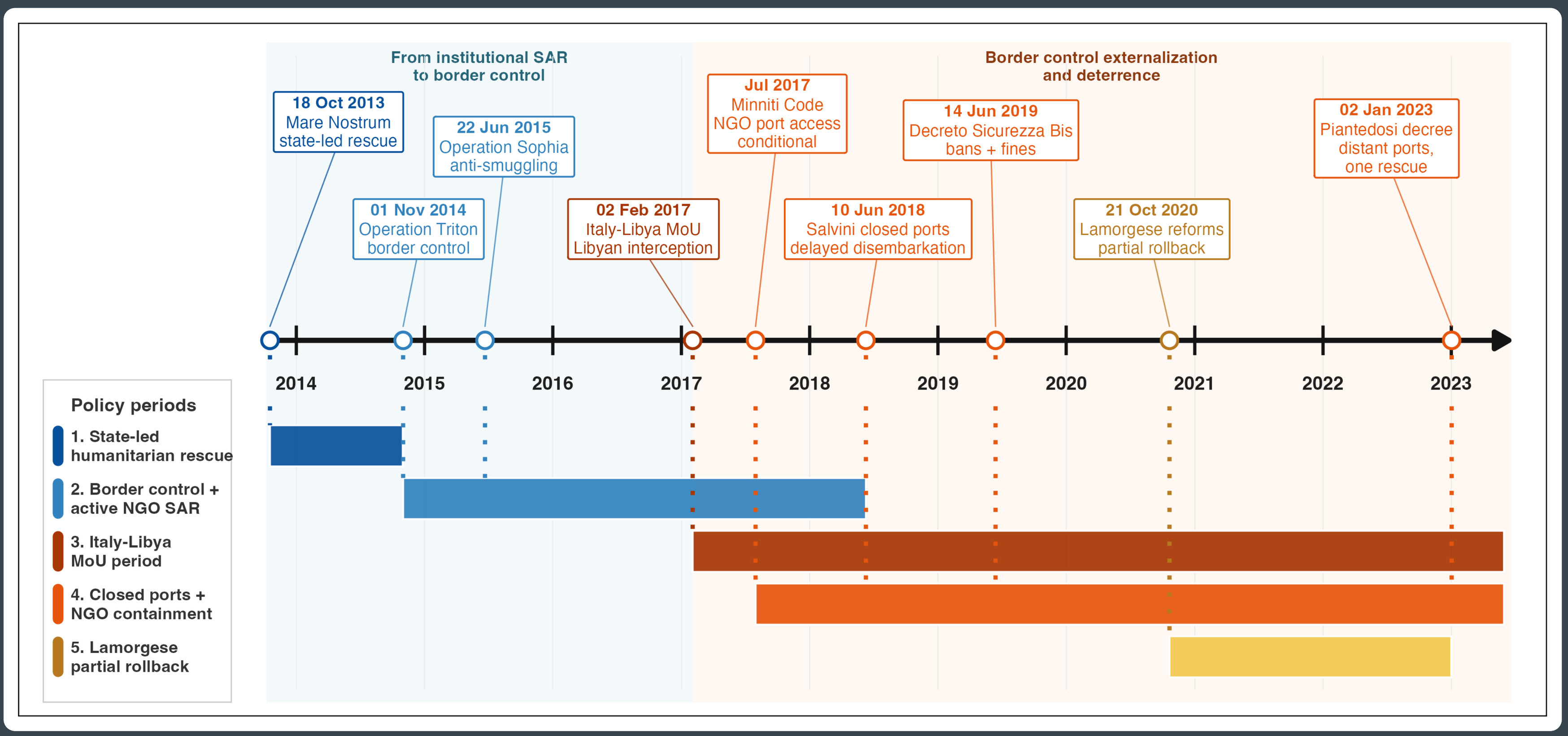
1 Motivation: risk at sea

The Central Mediterranean route (**CMR**) is the world's deadliest migration route: **20,368** recorded deaths in the 2014–2023 corridor sample alone. In 2017 the Italy–Libya Memorandum (**MoU**) turned EU policy from rescue toward interception, financing the Libyan Coast Guard and constraining NGO search-and-rescue (**SAR**). Crossings fell, but whether each crossing became *safer*, or merely less rescued, stayed open. The key margin is therefore **risk per crossing**.

2 Mechanism: SAR and risk moderation

Sea state acts through two channels. Rough seas **deter departures**, lowering recorded deaths; but conditional on departing, they raise the chance of distress, capsize and death. **SAR moderates this second, per-crossing channel.** As European state-led and NGO-led SAR withdrew from 2017 onward, that buffer should weaken, so the same seas should map into **more** recorded deaths. The prediction is a positive post-MoU shift in the weather–mortality slope.

Fig 1 | From rescue to interception

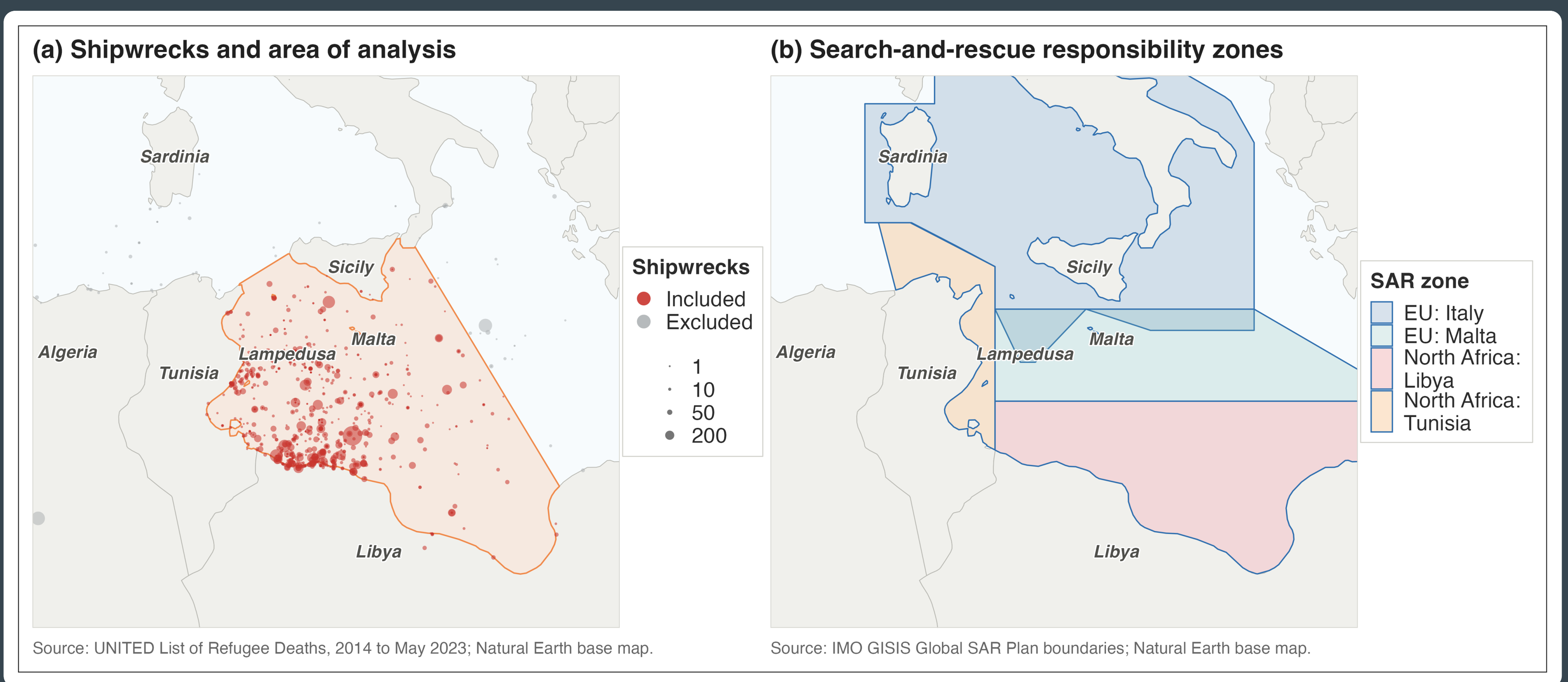


Central Mediterranean rescue and border-control policy, 2013–2023.

3 Data: daily route-level time series

A daily time series for the corridor, **January 2014–May 2023**, combining ERA5 wave height (**SWH**), **UNITED** and **IOM** recorded deaths, Frontex interceptions, and Libyan and Tunisian coast-guard pullbacks. The main regressor is the lagged five-day mean of wave height; the regime split is dated to the MoU signing on **2 February 2017**. This links sea state, mortality and crossing proxies on a common daily calendar.

Fig 2 | The corridor and SAR zones



Area of analysis, UNITED deadly incidents, and the Italian, Maltese, Libyan and Tunisian search-and-rescue responsibility zones.

4 Identification: exogenous sea-state variation

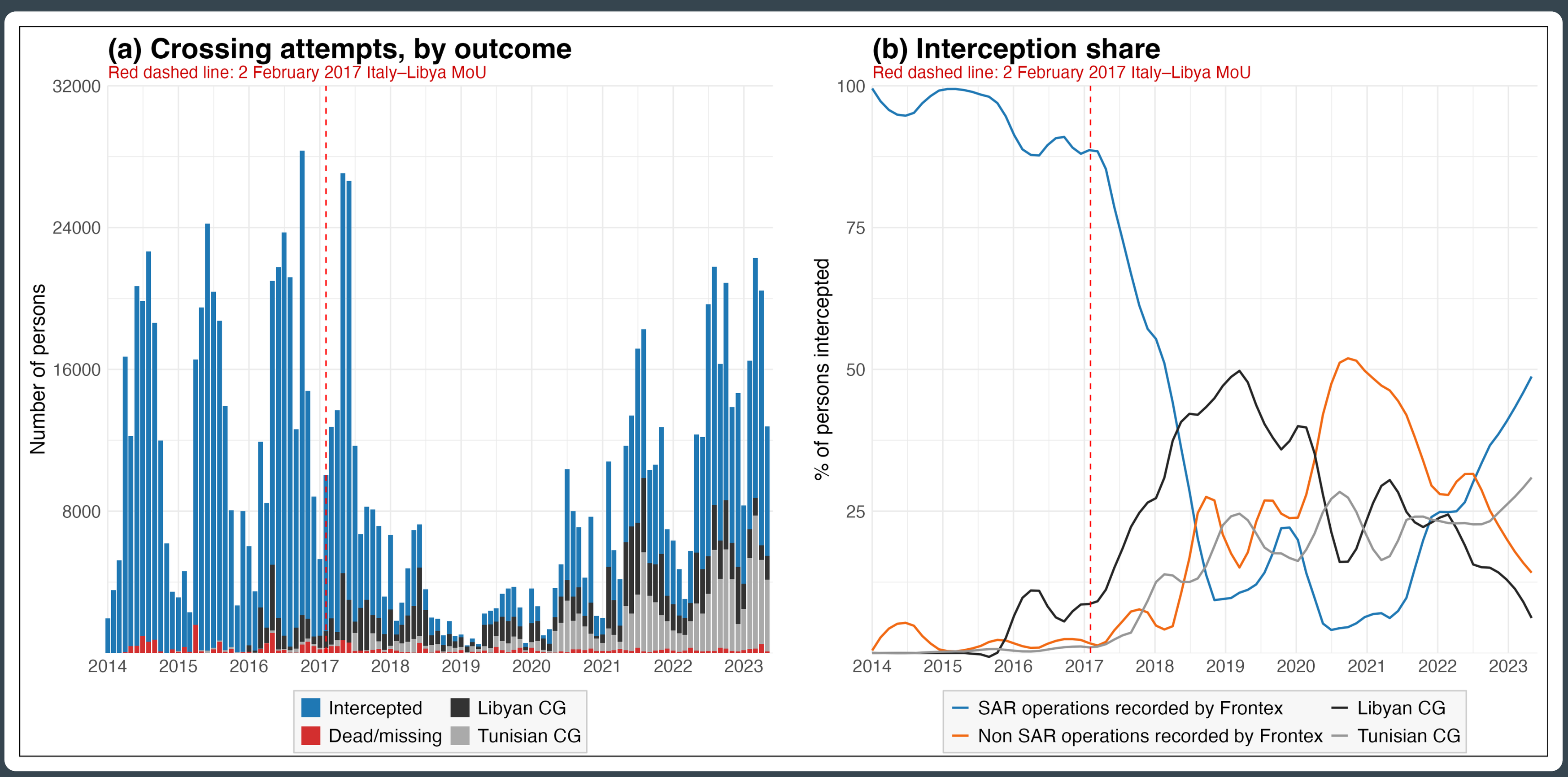
Day-to-day sea state is **plausibly exogenous** to migration policy: the same rough weather recurs before and after 2017, whoever governs rescue. Conditional on month–year fixed effects, we ask how the slope from lagged SWH to recorded deaths **shifts** across the MoU. **Main limits**: no untreated comparison route and incomplete records; observed boat composition is checked, but unobserved timing, route choice, boat quality and crowding still limit attribution to SAR alone.

The model

$$\mathbb{E}[D_t] = \exp\left(\alpha_{m(t)} + \beta_1 s_t + \beta_3\, s_t \times \text{Post}_t\right)$$

A negative-binomial and Poisson **count model** with month–year fixed effects; s_t is the lagged 5-day mean wave height and $\text{Post}_t = 1$ after the MoU. The estimand is β_3: how much the wave-height-to-deaths slope shifts after 2017.

Fig 3 | Crossings and interceptions



Persons attempting the crossing by outcome, and the share intercepted by operation type, 2014–2023.

5 Result: rough seas became riskier

In the primary UNITED negative-binomial model, rougher seas predicted about **94% fewer** recorded deaths before the MoU. The post-MoU interaction adds $+3.63$, so the implied post-MoU slope is $+0.80$. With the log link, $\exp(0.80) = 2.23$: expected deaths are **2.23 times as high** per additional metre of lagged SWH.

Parameter	Estimate	Interpretation
Pre-MoU slope β_1	-2.83^{***}	rough seas lower deaths
Post-MoU shift β_3	$+3.63^{***}$	adds to pre-MoU SWH effect
Implied post-MoU $\beta_1 + \beta_3$	$+0.80^{*}$	rough seas increase deaths by $2.23\times$ per +1m SWH

UNITED NegBin, month–year FE, Newey–West SEs (lag 14). Estimates are log slopes.

The shift holds in both registries and both count models, and survives controls for crossing volume, boat composition and Libyan conflict.

6 Contribution of the research

This thesis estimates how the **per-crossing risk of death** on the CMR changed after the 2017 MoU. It asks whether sea-state hazard mapped into a different mortality gradient under the post-MoU period. The point is not only whether deterrence reduced departures or arrivals, but whether crossings became more dangerous.

The evidence is consistent with SAR acting as a **safety buffer**: after the MoU, rough seas became more strongly associated with recorded deaths, *even as crossings declined*. The thesis proposes a new way to measure the human cost of this policy approach.

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